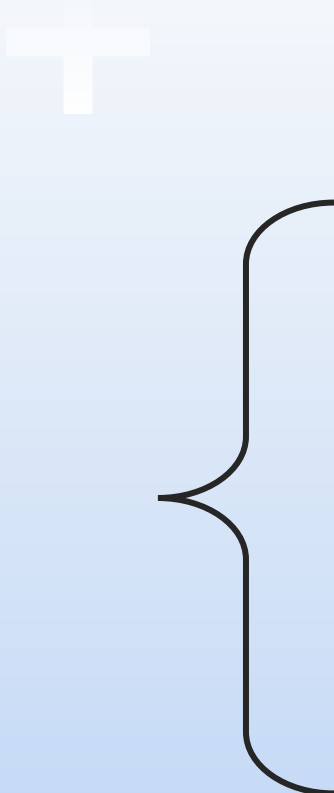


Cyclistic Bike-Share Analysis

How Annual Members and Casual Riders Use Cyclistic Bikes Differently

Agenda

- 
- Compare trip volume, ride duration, timing, bike type, and station use for annual members and casual riders
 - Use the findings to recommend ways to convert casual riders into annual members

Stakeholders:

Lily Moreno, Marketing Analytics Team, and Cyclistic Executive Team

Analytical Goals

Why

Understand how annual members and casual riders use Cyclistic differently

How

Compare trip volume, ride duration, timing, bike choice, and station activity



4

Data Preparation

Original trip records

6,115,982

Clean records analyzed

5,951,516

Removed duplicates, missing values, invalid rider types, and trips outside the valid range of 1 to 1,440 minutes.



Members Accounted for Most Rides

Members

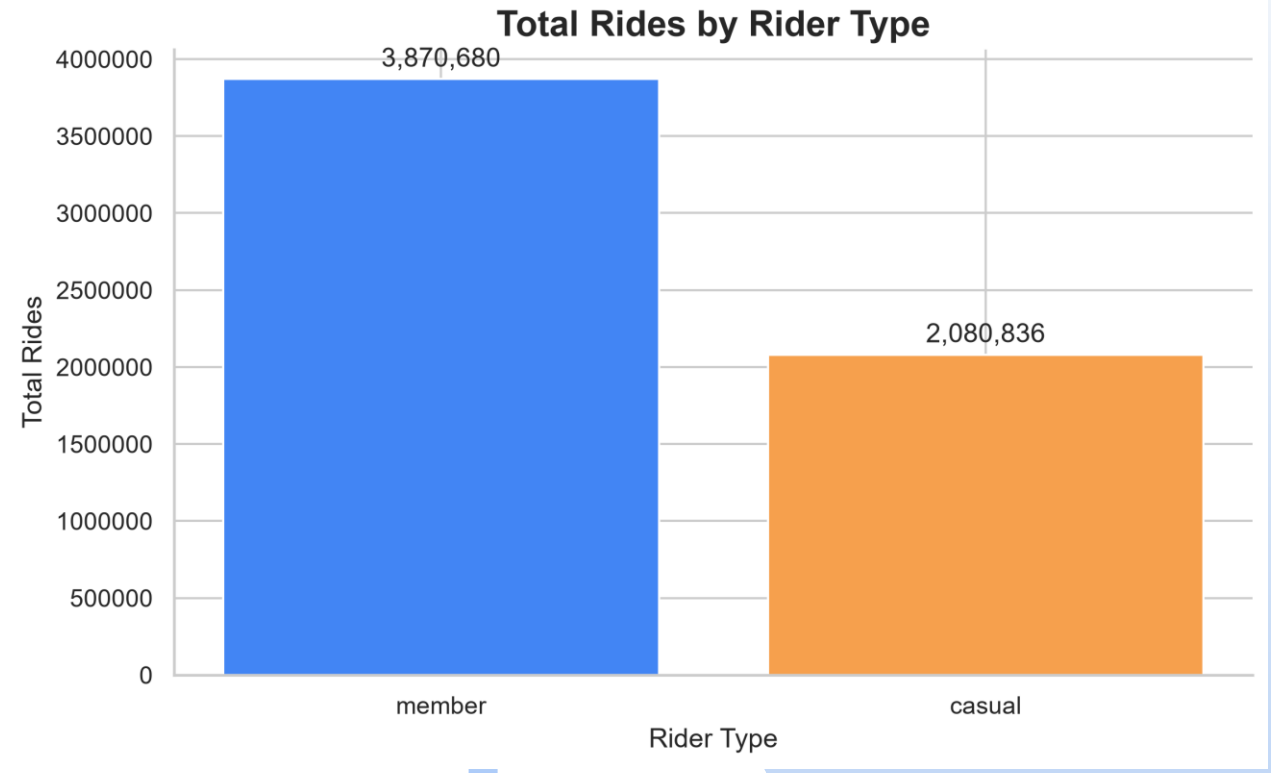
3,870,680 rides

65.04% of total rides

Casual Riders

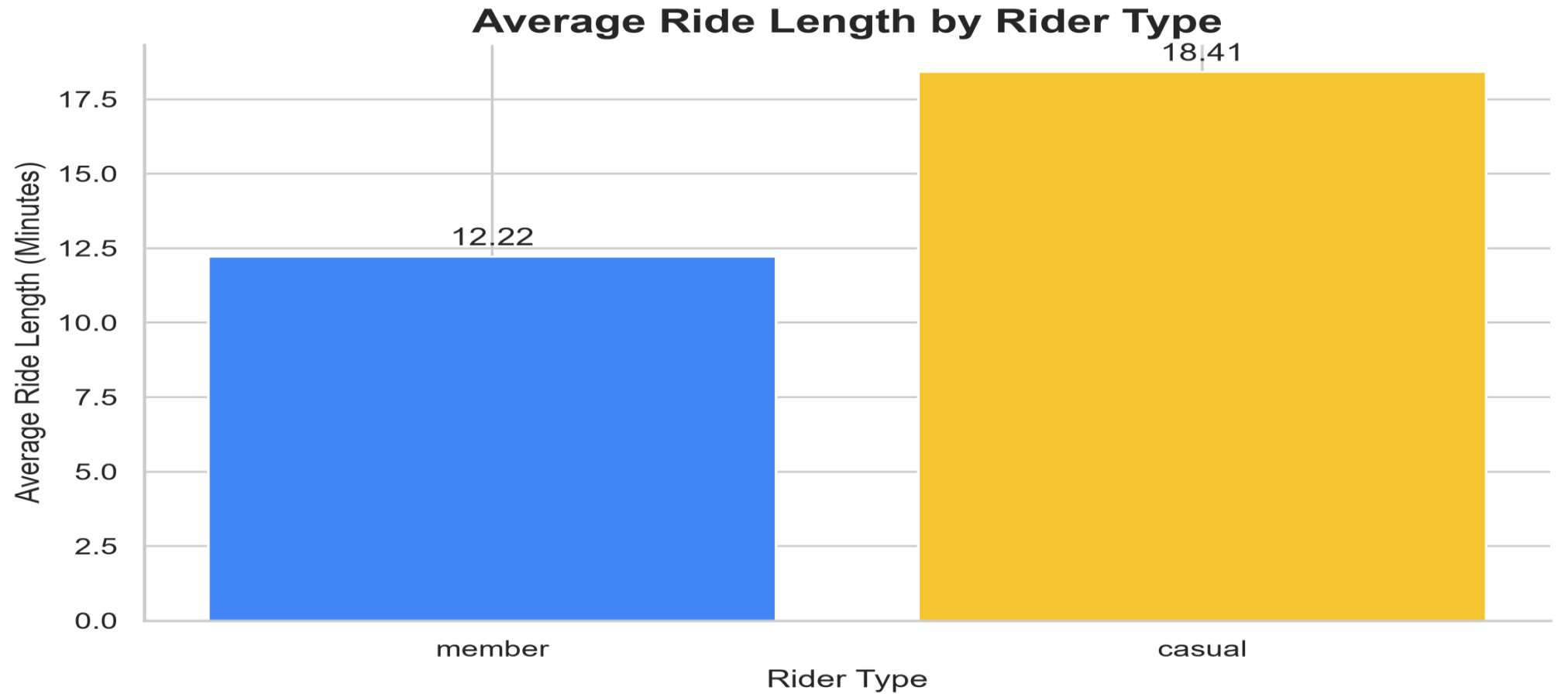
2,080,836 rides

34.96% of total rides



Members generated nearly two-thirds of all rides during the analysis period

Casual Riders Took Longer Trips

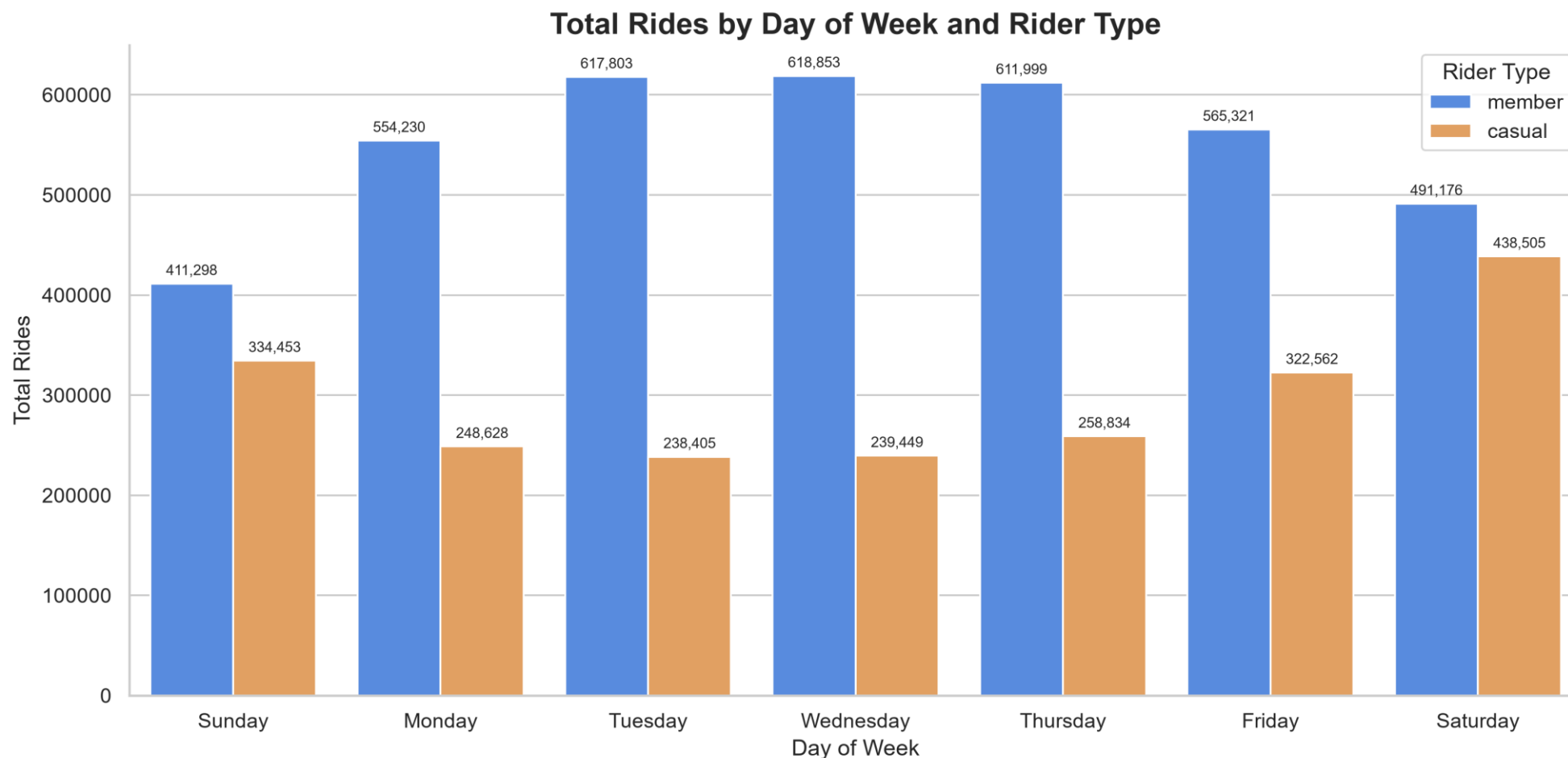


Casual riders
18.41 minutes

Members
12.22 minutes

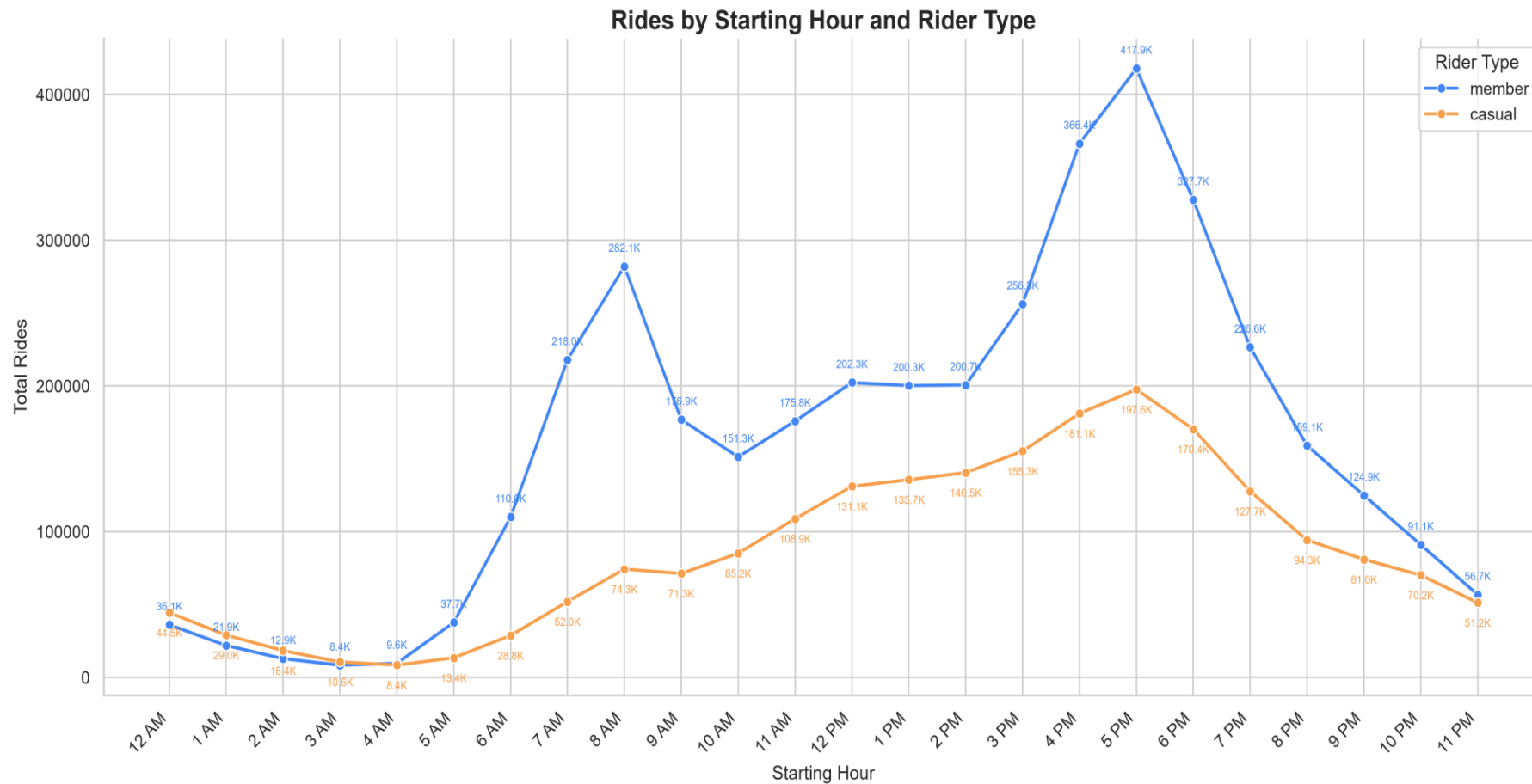
Casual rides averaged 6.19 minutes longer, about 51% above the member average

Members Showed Stronger Weekday Usage



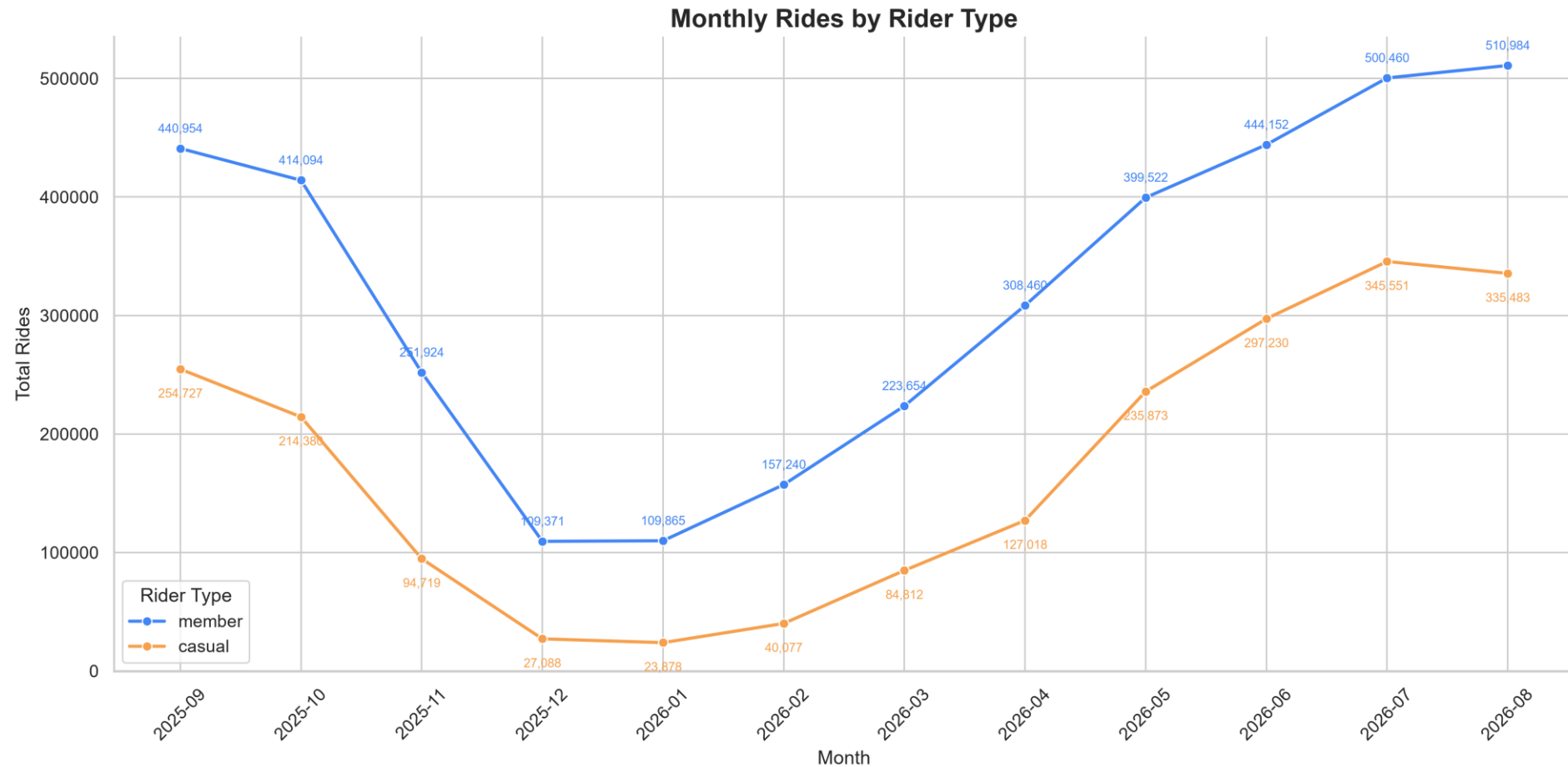
Members peaked on Wednesday with 618,853 rides, while casual riders peaked on Saturday with 438,505 rides

Members Showed Clear Commuting-Hour Peaks



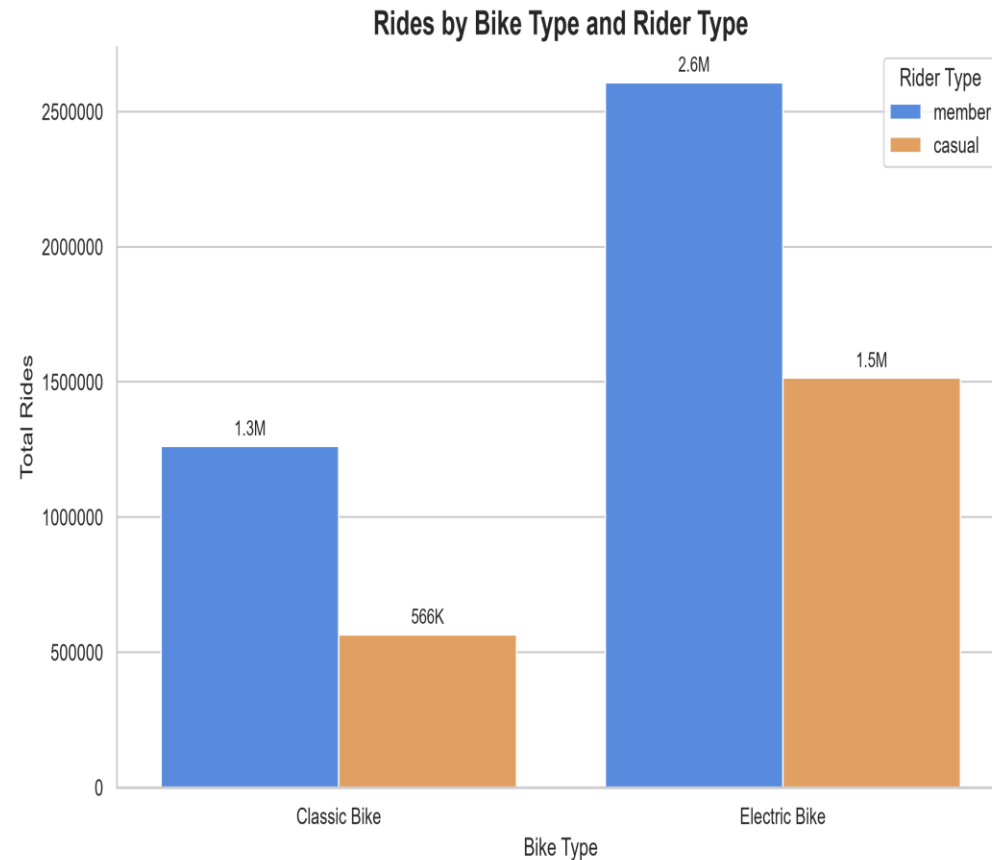
Member rides peaked near 8:00 AM and 5:00 PM, consistent with typical commuting hours

Ridership Followed a Seasonal Pattern



Ridership declined during winter and increased through spring and summer. Member rides peaked in August, while casual rides peaked in July

Electric Bikes Were Preferred by Both Groups

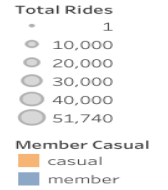
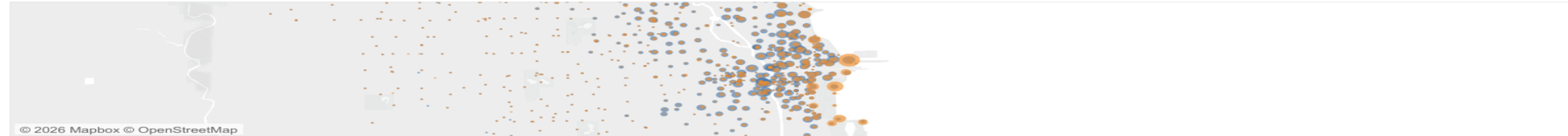


Electric bikes generated more rides than classic bikes among both members and casual riders

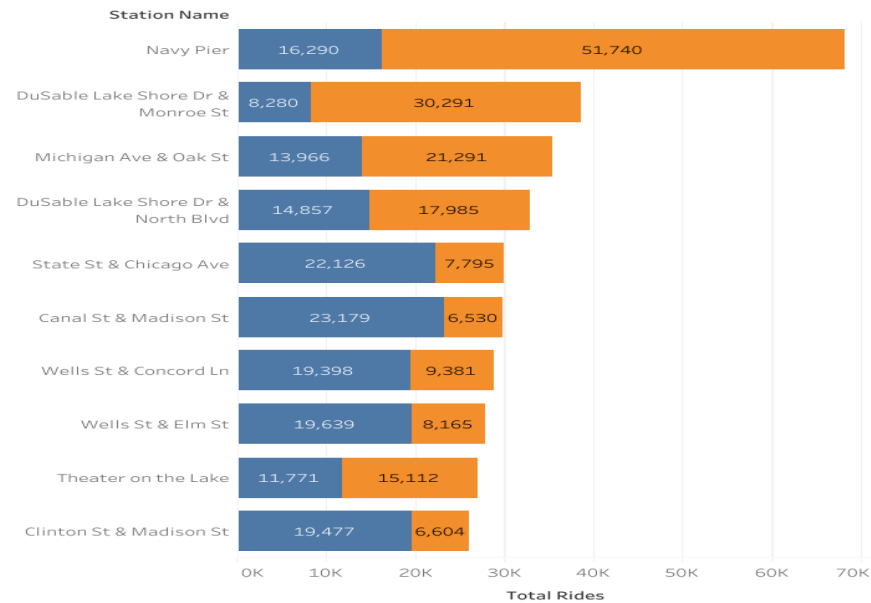
Navy Pier Led Station Activity

Station Analysis

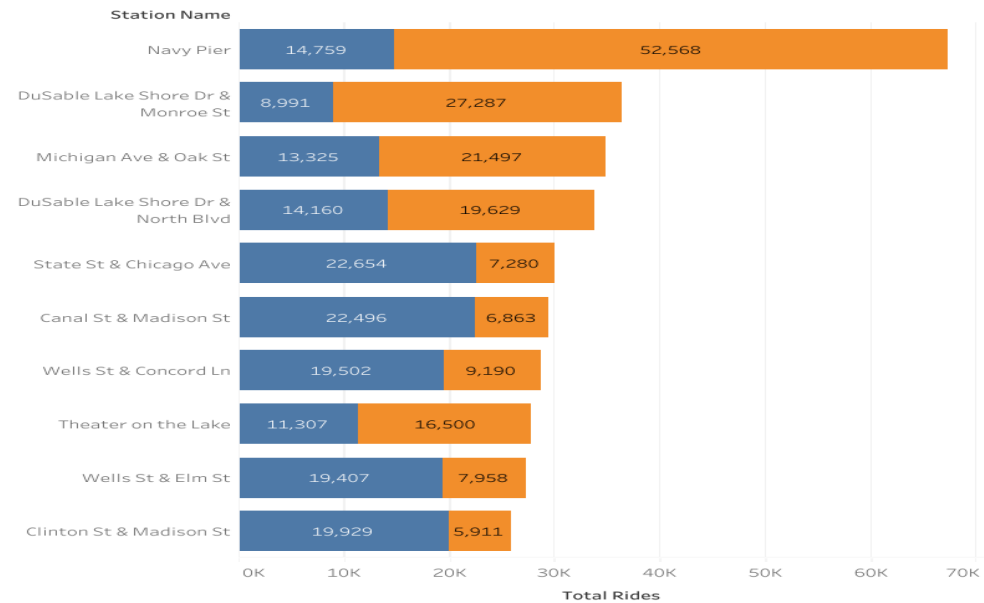
Station Popularity Map



Top Starting Stations



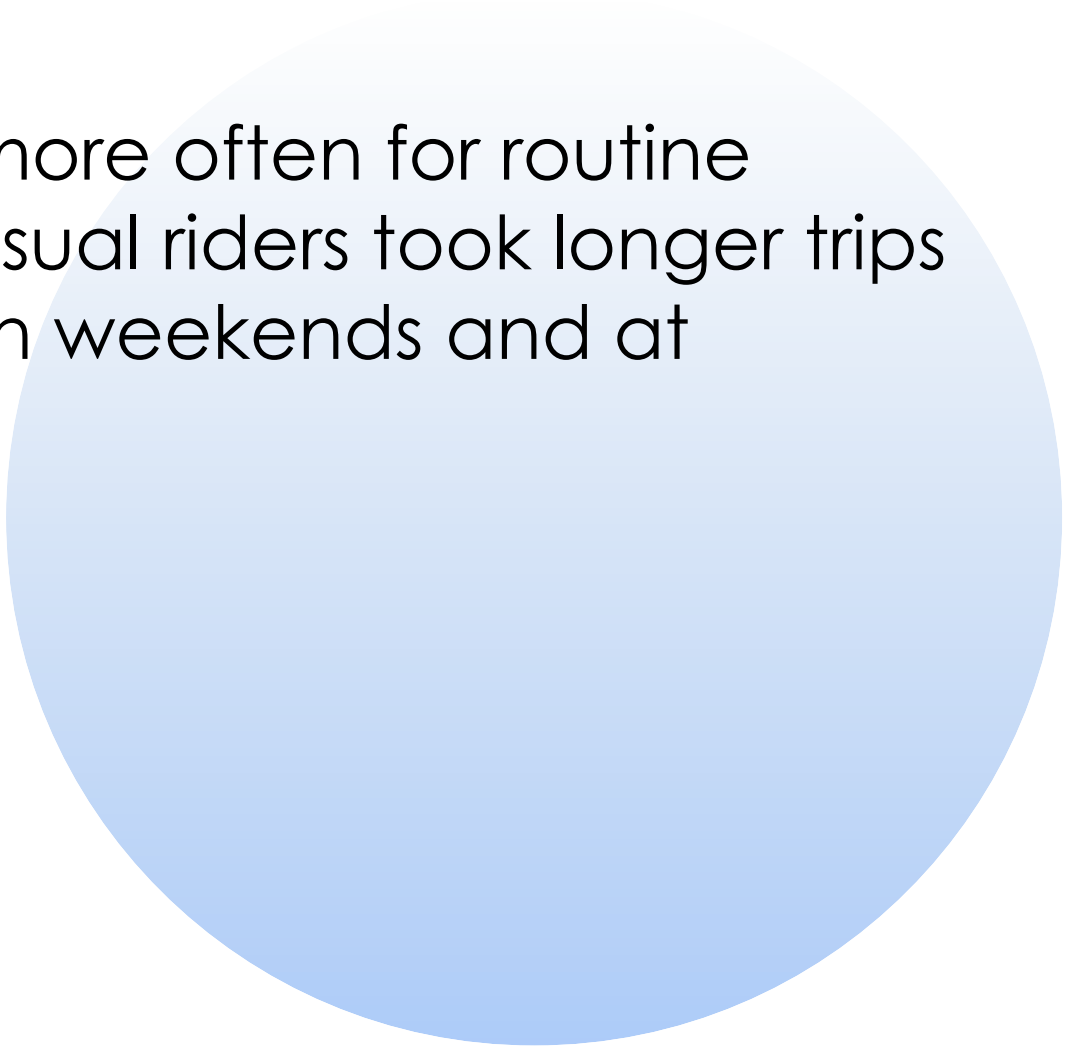
Top Ending Stations



Navy Pier recorded 68,030 starting rides and 67,327 ending rides, driven primarily by casual riders

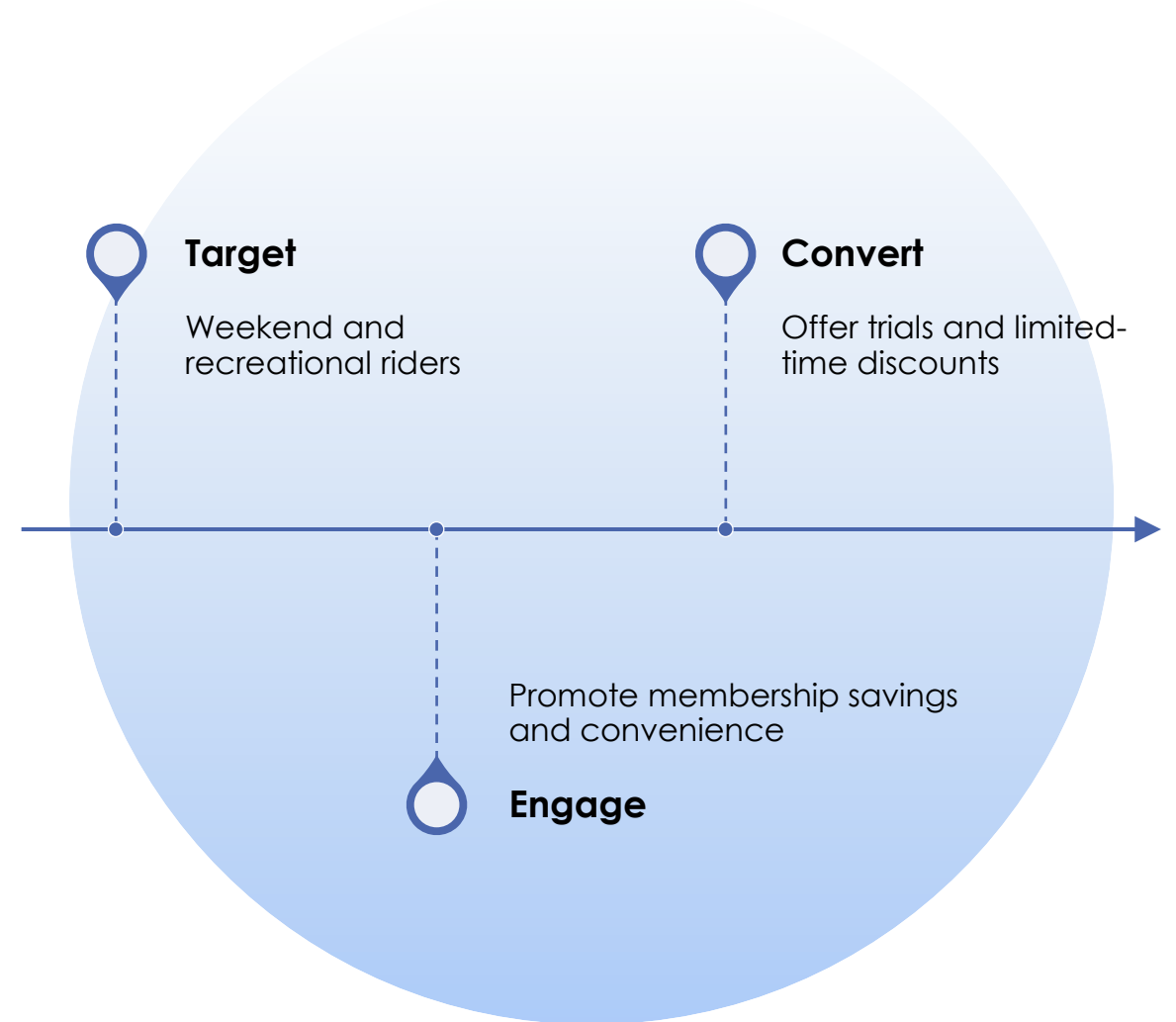
Summary

Members used Cyclistic more often for routine weekday travel, while casual riders took longer trips and rode more heavily on weekends and at recreational stations



Casual Rider Conversion Strategy

- Reach casual riders at high-traffic recreational stations
- Emphasize membership savings and convenience in digital advertising
- Offer trial memberships during peak riding months



Campaign Implementation Plan

Audience targeting

Identify casual rider segments by location, day, and season

Pilot and measurement

Test membership offers and track conversion rates

Campaign expansion

Scale the messages and offers that produce the strongest results

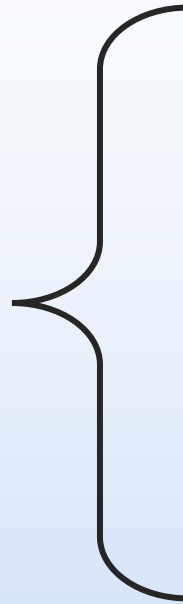
Milestone	Quarter
Audience segmentation	Q1
Pilot campaign	Q2
Performance evaluation	Q3
Campaign expansion	Q4

Opportunities for Further Analysis



- Include demographic data to better understand casual riders
- Analyze membership pricing and purchase history to estimate conversion value
- Track marketing exposure and membership conversions to measure campaign effectiveness

Interactive Dashboards and Project Files



[Tableau Public: Cyclistic Station Analysis](#)

[Looker Studio: Cyclistic Bike-Share Dashboard](#)

[GitHub: Full Case Study, SQL, Python, and Data Files](#)

Thank You

Denae Camacho

Google Data Analytics Capstone Project 2026